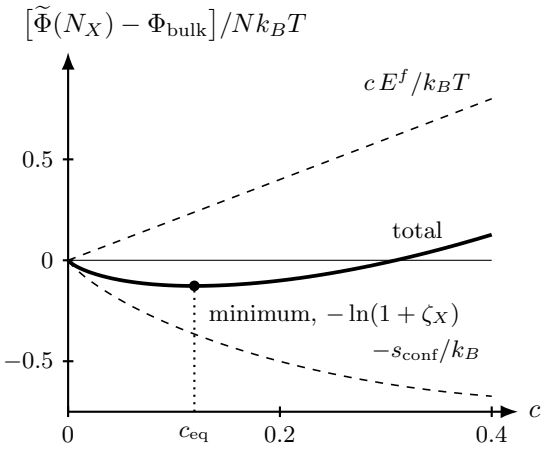
(a) $E^f = 2k_BT$, $g_X = 1$



(b) $g_X = 1$

